Maths Review for Machine Learning and Data Science
Topic 3: Calculus

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(with thanks to Geoff Coates for examples and diagrams on slides 5-10)

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BY NC ND

Introduction

Maths Review for Machine Learning and Data Science:

1. Functions and Notation
2. Linear Algebra
   - Vectors
   - Matrices
   - Converting equations to vector and matrix form
3. **Calculus**
   - Differentiation
   - Partial derivatives
   - Matrix calculus
4. Optimisation
   - Minima and maxima
   - Argmin and Argmax
   - Convex functions

Introduction

A few important notes:

- These workshops will review the fundamentals of each topics. Suggestions for further reading are given at the end those wishing to review or learn more advanced content. Go
- If you have any questions during the workshop, please raise your hand and ask! If you would like to discuss any of these topics in more detail, one-on-one, there is information at the end about accessing our free drop-in and consultation services. Go

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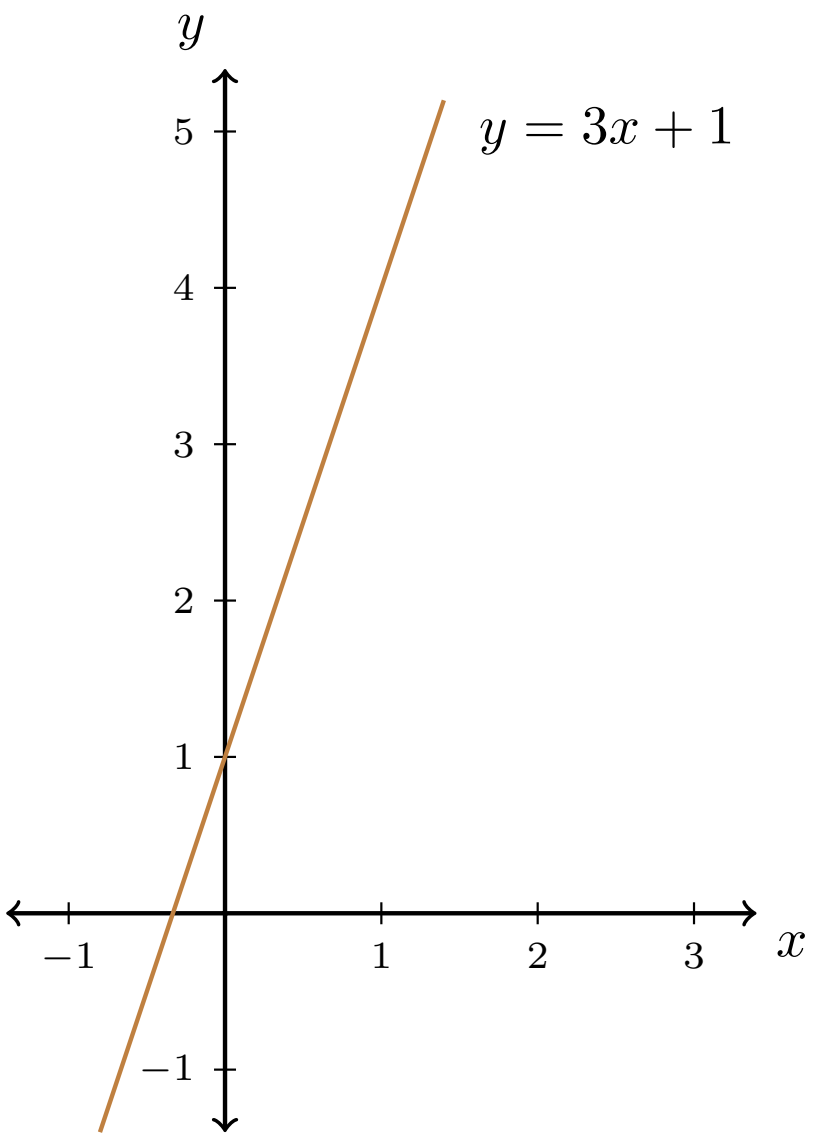
The gradient of a line

For a linear function,

$$y = mx + c,$$

the *gradient* is the slope of the line, given by the coefficient m.

For example, the gradient of the function $y = 3x + 1$ is 3:



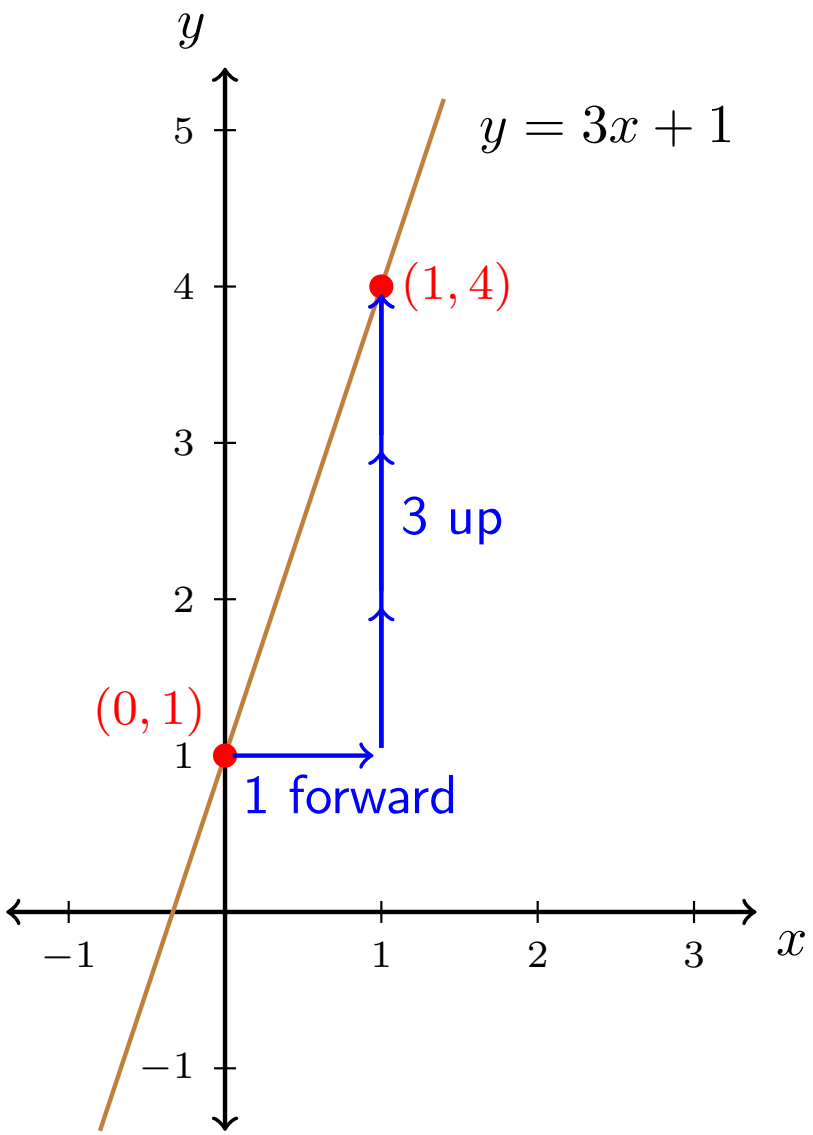
The gradient of a line

We can take calculate the gradient by finding the 'rise over run', that is:

$$\frac{\text{rise}}{\text{run}} = \frac{\text{change in } y}{\text{change in } x} = \frac{\Delta y}{\Delta x}.$$

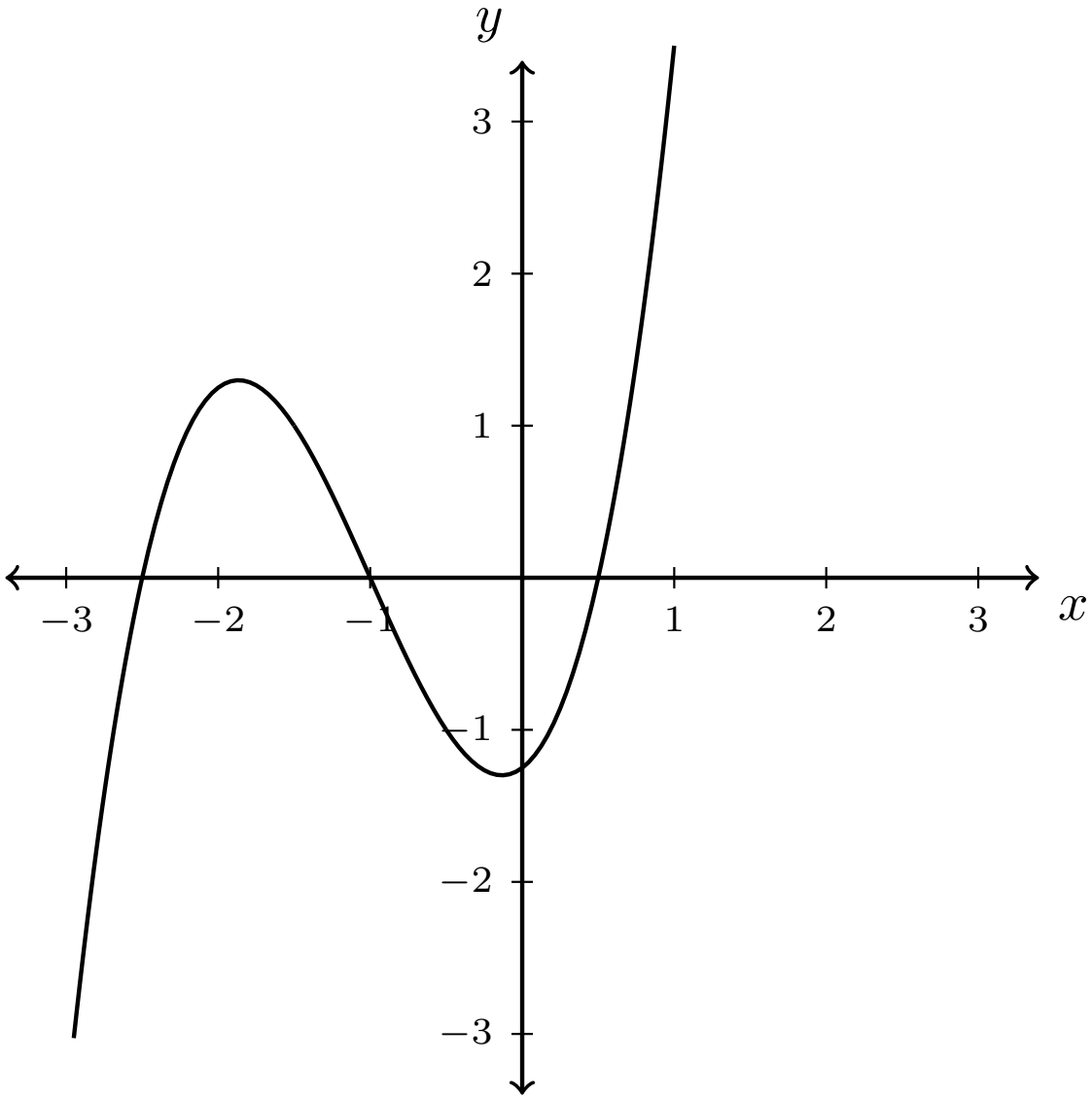
In our example, we find that

$$\frac{\Delta y}{\Delta x} = \frac{3}{1} = 3.$$



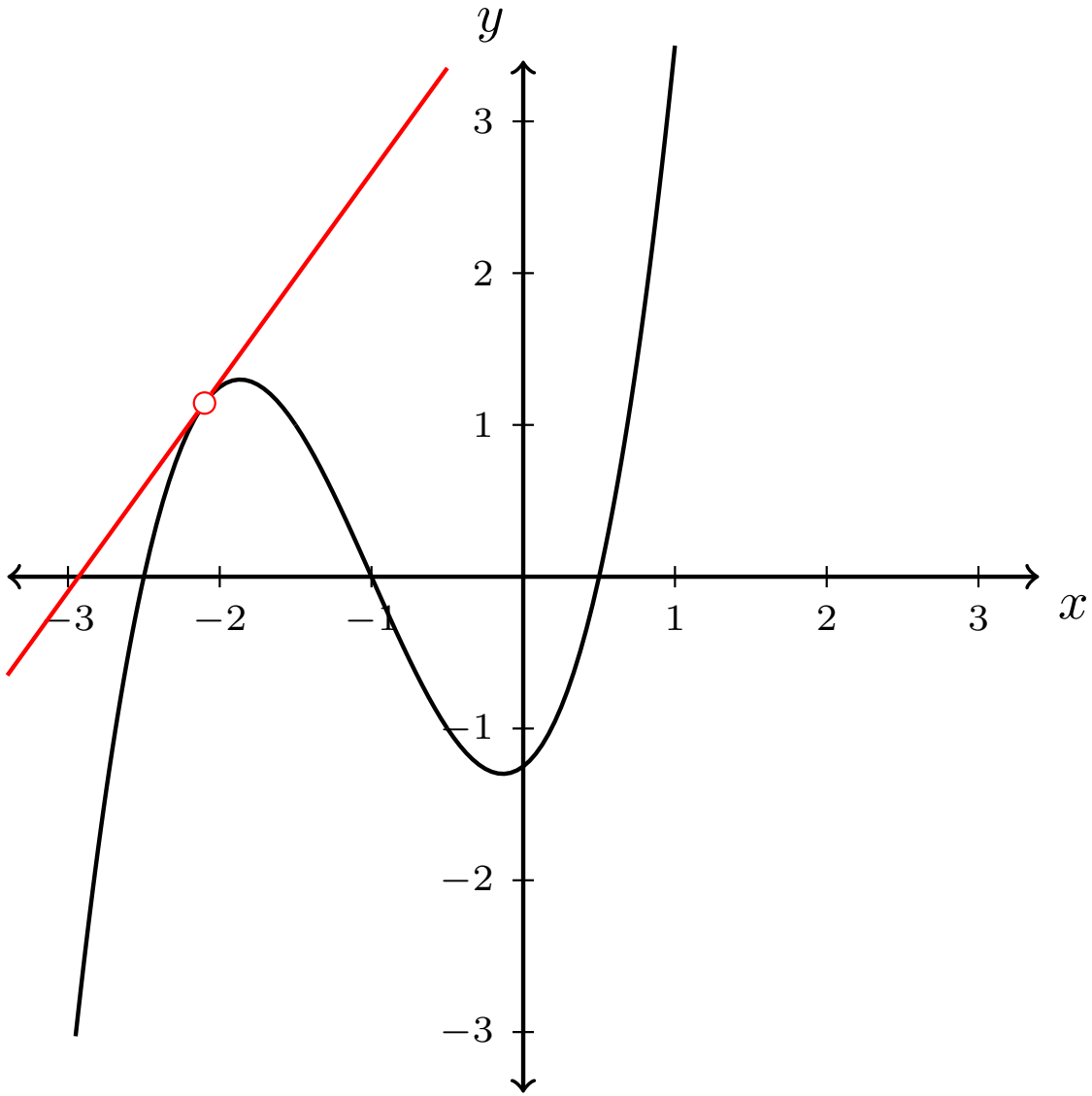
The derivative of a function

For non-linear functions, the line does not have a consistent slope.



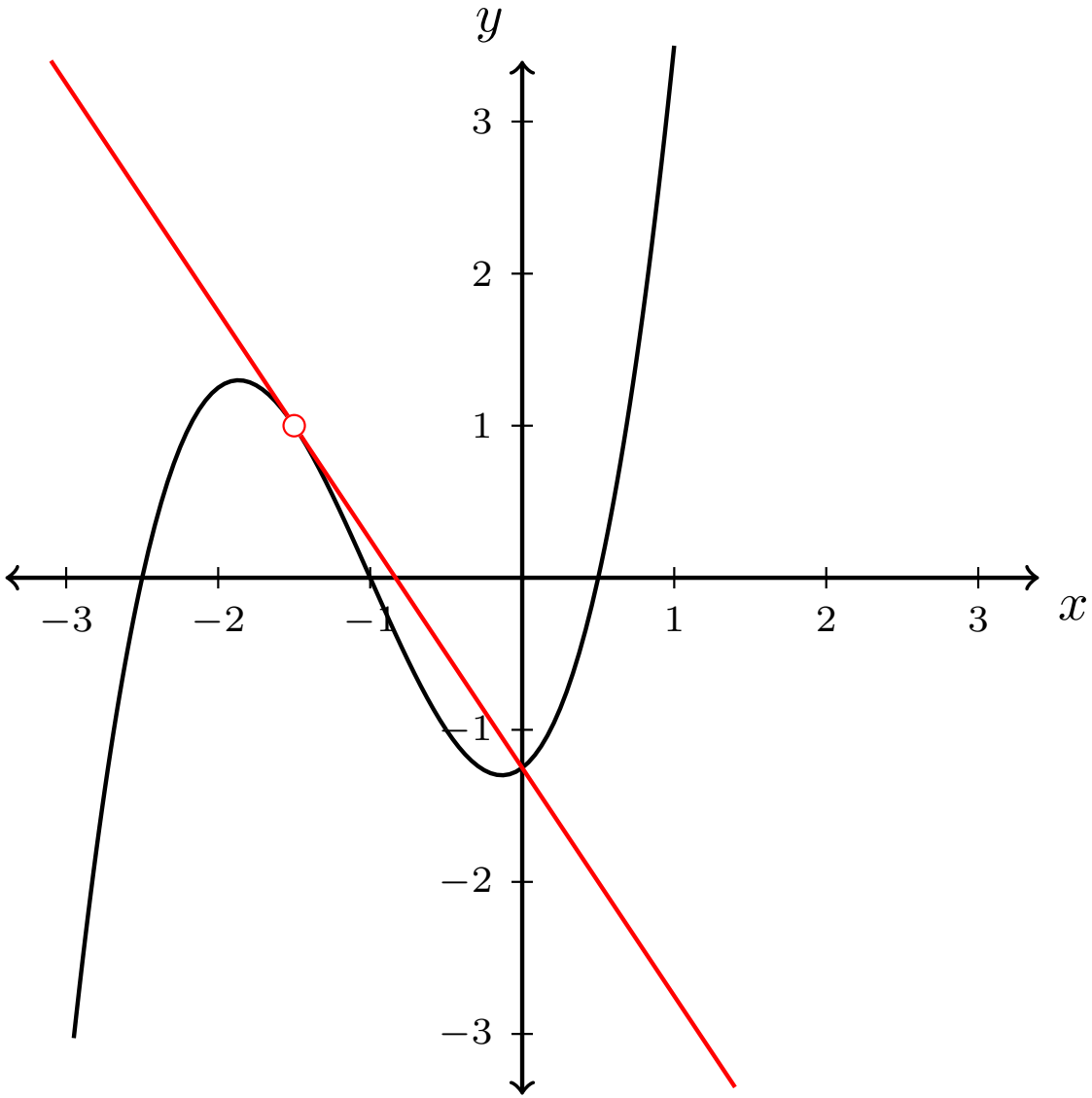
The derivative of a function

However, we can find a *tangent* line to the function for every point in the domain:



The derivative of a function

This tangent line will be different at different points.



The derivative of a function

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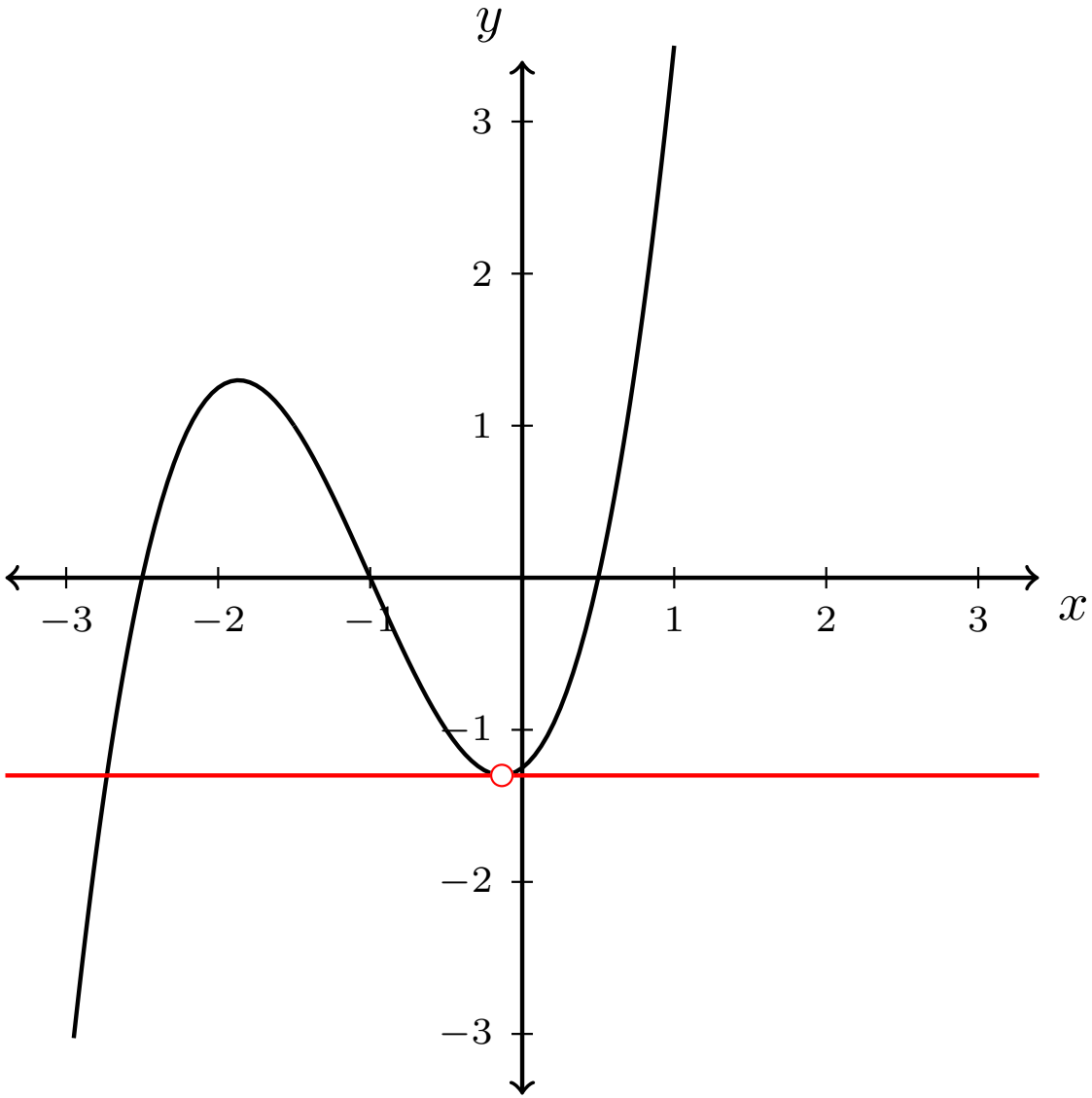


The derivative of a function

We can find the slope of the tangent line, or the *instantaneous rate of change* of a function by taking its *derivative*.

We denote the derivative a few different ways.

Let's say we have a function $f(x)$. We might write the derivative as:

$$\frac{d}{dx}f(x) \qquad \frac{dy}{dx}$$

$$\frac{df}{dx} \qquad f'(x)$$

The derivative of a function

Returning to our example of a linear function, $y = 3x + 1$, we might write the derivative as

$$\frac{d}{dx}(3x+1) \quad \text{or simply as} \quad \frac{dy}{dx}.$$

If we rewrite our function as $f(x) = 3x + 1$, then we might write the derivative as

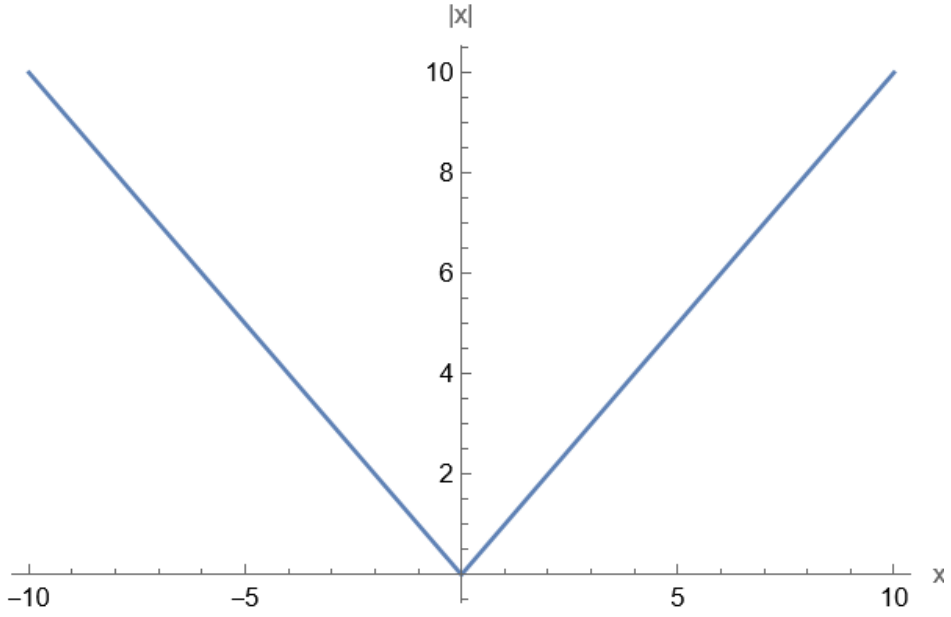
$$\frac{df}{dx} \quad \text{or} \quad f'(x).$$

The derivative of a function

To be considered *differentiable*, a function must be differentiable at every point on its domain. Not every function is differentiable.

To be differentiable, a function must be *continuous*, that is, it mustn't have any breaks.

However, some continuous functions are not differentiable. For example, functions with an angle, like the absolute value function.



A formal definition

Formally, a function f is said to be differentiable at a point a on its domain if the limit L exists, where

$$L = \lim_{b \to 0} \frac{f(a+b) - f(a)}{b}.$$

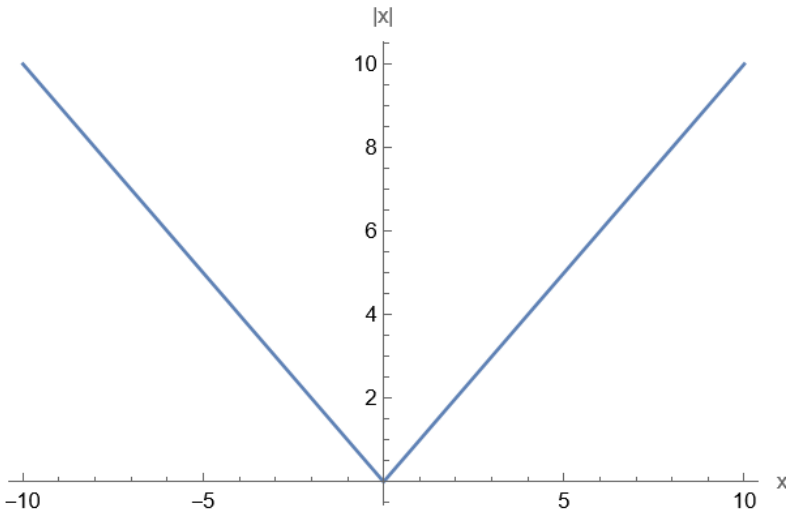
If f is differentiable at a, then $f'(a) = L$.

Returning the the example of the absolute value, we can see that the limit at the point $a = 0$, as b approaches zero from the left,

$$\lim_{b \to 0^-} \frac{f(b)}{b} = \frac{-b}{b} = -1.$$

While as b approaches zero from the right,

$$\lim_{b \to 0^+} \frac{f(b)}{b} = \frac{b}{b} = 1.$$



Some common derivatives

Rules:

$f(x)$	$f'(x)$
c	0
cx	c
x^n	nx^{n-1}
$\ln x$	$\frac{1}{x}$
e^x	e^x
$\log_a x$	$\frac{1}{x \ln a}$
a^x	$a^x \ln a$
$\sin(x)$	$\cos(x)$
$\cos(x)$	$-\sin(x)$

Examples:

$f(x)$	$f'(x)$
2	0
$2x$	2
x^2	$2x$

$\log_2 x$	$\frac{1}{x \ln 2}$
2^x	$2^x \ln 2$

Some differentiation rules

	Function	Derivative
Constant multiple rule	$cf(x)$	$cf'(x)$
Sum rule	$f(x) + g(x)$	$f'(x) + g'(x)$
Difference rule	$f(x) - g(x)$	$f'(x) - g'(x)$
Product rule	$f(x) \cdot g(x)$	$f'(x) \cdot g(x) + g'(x) \cdot f(x)$
Quotient rule	$\frac{f(x)}{g(x)}$	$\frac{f'(x) \cdot g(x) - g'(x) \cdot f(x)}{[g(x)]^2}$
Chain rule	$f(g(x))$	$f'(g(x)) \cdot g'(x)$

Some differentiation rules

Example 1:

$$\frac{d}{dx}\left(x^2 \ln x\right) = \frac{dx^2}{dx} \cdot \ln x + x^2 \cdot \frac{d \ln x}{dx} = 2x \ln x + x$$

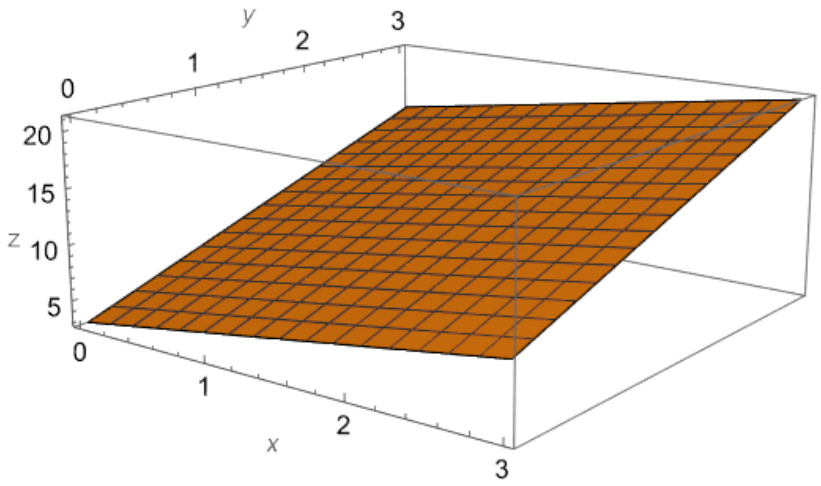
Example 2:

$$\begin{aligned}
\frac{d}{dx}\left(\sin(4x) - \sqrt{x}\right) &= \frac{d\sin(y)}{dy} \cdot \frac{d4x}{dx} - \frac{d\sqrt{x}}{dx} && (\text{let } y = 4x) \\
&= 4\cos(y) - \frac{1}{2}x^{-1/2} \\
&= 4\cos(4x) - \frac{1}{2\sqrt{x}} && (\text{substitute } y = 4x)
\end{aligned}$$

Partial Derivatives

When we have a function of more than two variables, we may have different 'slopes' in different dimensions.
Consider the example, from the Topic 1 workshop, of the linear function

$$f(x, y) = 2x + 4y + 3 :$$



As we move in the direction of the x-axis, we have a slope of 2, while in the y-axis the slope is 4.

Partial Derivatives

In the case of the linear function, it is fairly clear what these indivisual slopes are. What do we do in the case of non-linear functions?

Partial derivatives are derivatives of multi-variable taken with respect to only one of the variables, with all other variables held constant.

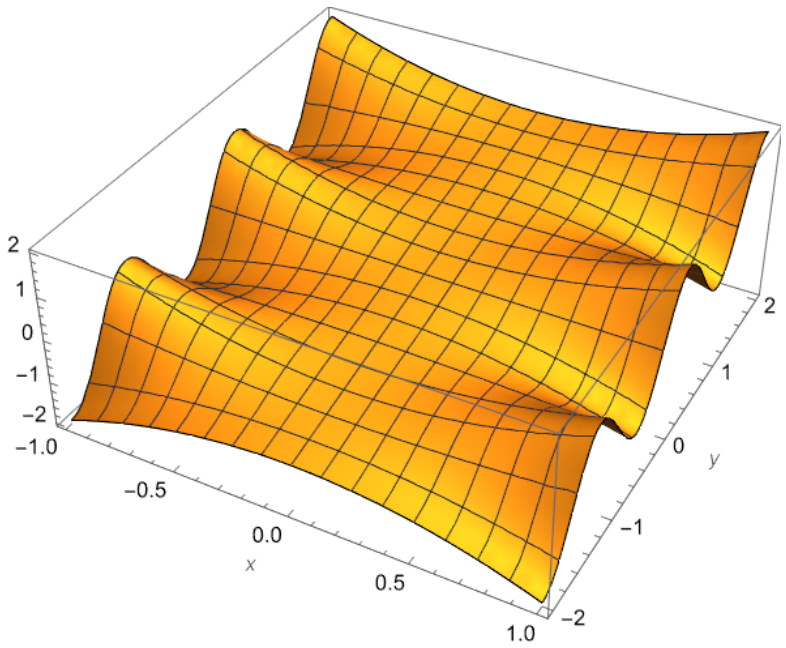
For example, for our linear function $f(x, y) = 2x + 4y + 3$,

$$\frac{\partial f}{\partial x} = 2 \quad \text{and} \quad \frac{\partial f}{\partial y} = 4.$$

Partial Derivatives

Recall, *Partial derivatives* are derivatives of multi-variable taken with respect to only one of the variables, with all other variables held constant.

For a more complex example, consider the function $g(x, y) = x^2 \sin(4y) + \frac{y}{2}$:

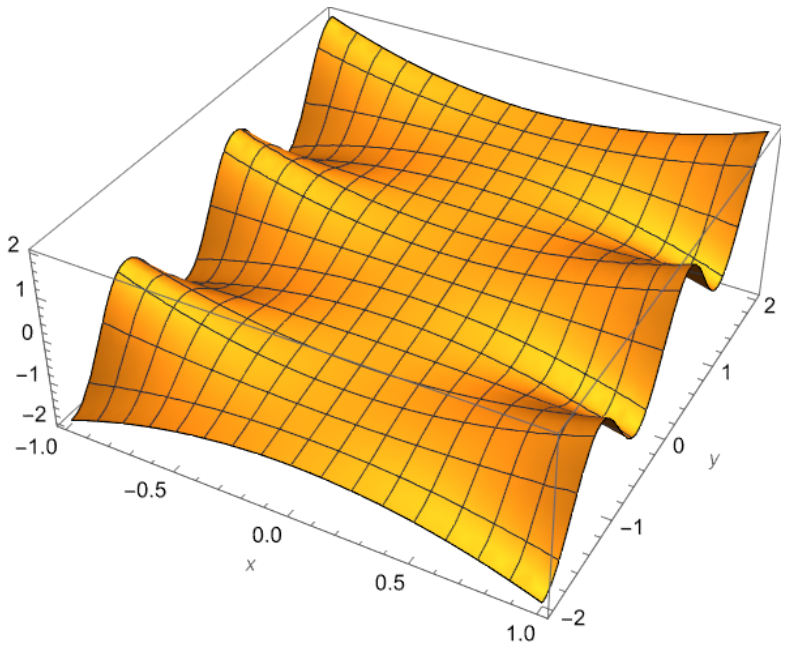


$$\begin{aligned}\frac{\partial g}{\partial x} &= \frac{\partial}{\partial x}\left(x^2 \sin(4y) + \frac{y}{2}\right) \\ &= \frac{\partial}{\partial x}\left(x^2 \sin(4y)\right) + \frac{\partial}{\partial x}\left(\frac{y}{2}\right) \\ &= 2x \sin(4y).\end{aligned}$$

Partial Derivatives

Recall, *Partial derivatives* are derivatives of multi-variable taken with respect to only one of the variables, with all other variables held constant.

For a more complex example, consider the function $g(x, y) = x^2 \sin(4y) + \frac{y}{2}$:



$$\begin{aligned}\frac{\partial g}{\partial y} &= \frac{\partial}{\partial y}\left(x^2 \sin(4y) + \frac{y}{2}\right) \\ &= \frac{\partial}{\partial y}\left(x^2 \sin(4y)\right) + \frac{\partial}{\partial y}\left(\frac{y}{2}\right) \\ &= 4x^2 \cos(4y) + \frac{1}{2}\end{aligned}$$

Differentiation of vector-valued functions

Functions with vector outputs have many of the same differentiation rules as real-valued functions. Let $\mathbf{f}$ and $\mathbf{g}$ be functions that map from some (scalar or vector) parameter x to a vector. Then we have:

	Function	Derivative
Constant multiple rule	$c\mathbf{f}(x)$	$c\mathbf{f}'(x)$
Sum rule	$\mathbf{f}(x) + \mathbf{g}(x)$	$\mathbf{f}'(x) + \mathbf{g}'(x)$
Difference rule	$\mathbf{f}(x) - \mathbf{g}(x)$	$\mathbf{f}'(x) - \mathbf{g}'(x)$
Dot product rule	$\mathbf{f}(x) \cdot \mathbf{g}(x)$	$\mathbf{f}'(x) \cdot \mathbf{g}(x) + \mathbf{f}(x) \cdot \mathbf{g}'(x)$
Chain rule	$\mathbf{f}(\mathbf{g}(x))$	$\mathbf{f}'(\mathbf{g}(x)) \cdot \mathbf{g}'(x)$

Differentiation of vector-valued functions

Let $\mathbf{x} \in \mathbb{R}^m$ and let $\mathbf{a} \in \mathbb{R}^m$ be a vector that is not a function of $\mathbf{x}$.

Then we have the following rules for partial derivatives:

$$\frac{\partial \mathbf{a}}{\partial \mathbf{x}} = \mathbf{0}_{n \times m} \quad \text{where } \mathbf{0}_{n \times m} \text{ is the } n \times m \text{ matrix of zeros}$$

$$\frac{\partial \mathbf{x}}{\partial \mathbf{x}} = \mathbf{I}_n \quad \text{where } \mathbf{I}_n \text{ is the } n \times n \text{ identity matrix}$$

$$\frac{\partial}{\partial \mathbf{x}}(\mathbf{a}^\top \mathbf{x}) = \frac{\partial}{\partial \mathbf{x}}(\mathbf{x}^\top \mathbf{a}) = \mathbf{a}$$

$$\frac{\partial}{\partial \mathbf{x}}(\mathbf{x}^\top \mathbf{x}) = 2\mathbf{x}$$

Differentiation of vector-valued functions

Additionally, let $A \in \mathbb{R}^{m \times n}$ be a matrix that is not a function of $\mathbf{x}$, then we have the following rules:

$$\frac{\partial}{\partial \mathbf{x}}(\mathbf{A}\mathbf{x}) = \mathbf{A}^\top \quad \text{and} \quad \frac{\partial}{\partial \mathbf{x}}(\mathbf{x}^\top \mathbf{A}) = \mathbf{A}.$$

Finally, let u and v be vector-valued functions of x, then we have:

$$\frac{\partial \mathbf{u}^\top}{\partial \mathbf{x}} = \left(\frac{\partial \mathbf{u}}{\partial \mathbf{x}}\right)^\top \quad \text{and} \quad \frac{\partial}{\partial \mathbf{x}}(\mathbf{u}^\top \mathbf{v}) = \frac{\partial \mathbf{u}}{\partial \mathbf{x}}\mathbf{v} + \frac{\partial \mathbf{v}}{\partial \mathbf{x}}\mathbf{u}.$$

Differentiation of vector-valued functions

Here is an example you might recall from a CITS5508 lecture.

We have:

$$\text{RSS}(\mathbf{X}, h_\theta) = (\mathbf{y} - \mathbf{X}\theta)^\top (\mathbf{y} - \mathbf{X}\theta),$$

where $\mathbf{y} \in \mathbb{R}^m$, $\mathbf{X} \in \mathbb{R}^{m \times n}$, and $\theta \in \mathbb{R}^n$.

Now we can find the partial derivative of RSS with respect to θ:

$$\begin{aligned}
\frac{\partial \text{RSS}}{\partial \theta} &= \frac{\partial}{\partial \theta} \left((\mathbf{y} - \mathbf{X}\theta)^\top (\mathbf{y} - \mathbf{X}\theta) \right) \\
&= 2 \left(\frac{\partial}{\partial \theta} (\mathbf{y} - \mathbf{X}\theta) \right) (\mathbf{y} - \mathbf{X}\theta) \\
&= 2 \left(\frac{\partial \mathbf{y}}{\partial \theta} - \frac{\partial}{\partial \theta} (\mathbf{X}\theta) \right) (\mathbf{y} - \mathbf{X}\theta) \\
&= 2 \left(\mathbf{0}_{n \times m} - \mathbf{X}^\top \right) (\mathbf{y} - \mathbf{X}\theta) \\
&= -2\mathbf{X}^\top (\mathbf{y} - \mathbf{X}\theta).
\end{aligned}$$

The gradient of a multi-variable function

The *gradient* of a function of multiple variables, or a function with a vector input, is the vector of partial derivatives.

For example, let's rewrite our linear function, $f : \mathbb{R}^2 \to \mathbb{R}$, as

$$f(\mathbf{x}) = 2x_1 + 4x_2 + 3.$$

Then we have the gradient:

$$\nabla_{\mathbf{x}} f = \begin{bmatrix} \frac{\partial f}{\partial x_1} \\ \frac{\partial f}{\partial x_2} \end{bmatrix} = \begin{bmatrix} 2 \\ 4 \end{bmatrix}.$$

You may also see this written as $\nabla f(\mathbf{x})$.

The gradient of a multi-variable function

The *gradient* of a function of multiple variables, or a function with a vector input, is the vector of partial derivatives.

In general, for a function, $f : \mathbb{R}^n \to \mathbb{R}$ we have the gradient:

$$\nabla_{\mathbf{x}} f = \begin{bmatrix} \frac{\partial f}{\partial x_1} \\ \frac{\partial f}{\partial x_2} \\ \vdots \\ \frac{\partial f}{\partial x_n} \end{bmatrix} \quad \text{or} \quad \nabla f(\mathbf{x}) = \begin{bmatrix} \frac{\partial f}{\partial x_1}(\mathbf{x}) \\ \frac{\partial f}{\partial x_2}(\mathbf{x}) \\ \vdots \\ \frac{\partial f}{\partial x_n}(\mathbf{x}) \end{bmatrix}$$

Types of matrix derivatives

There are various types of matrix derivatives, with the dimension of the derivative matrix depending on both the domain and codomain of the function:

		Output/Codomain		
		Scalar / $\mathbb{R}$	Vector / $\mathbb{R}^m$	Matrix / $\mathbb{R}^{n\times m}$
Input/Domain	Scalar / $\mathbb{R}$	Scalar / $\mathbb{R}$ (gradient)	Vector / $\mathbb{R}^{1\times m}$	Matrix / $\mathbb{R}^{m\times n}$
	Vector / $\mathbb{R}^n$	Vector / $\mathbb{R}^n$ (gradient)	Matrix / $\mathbb{R}^{m\times n}$ (Jacobian)	
	Matrix / $\mathbb{R}^{n\times m}$	Matrix / $\mathbb{R}^{n\times m}$ (gradient matrix)		

In CITS5508 you will likely only see those with a scalar codomain, the first column of this table.

The gradient matrix

The *gradient* of a scalar function with a matrix input, $f : \mathbb{R}^{n \times m} \to \mathbb{R}$ is the matrix of partial derivatives:

$$\nabla_{\mathbf{A}} f = \begin{bmatrix} \frac{\partial f}{\partial A_{11}} & \frac{\partial f}{\partial A_{21}} & \cdots & \frac{\partial f}{\partial A_{1m}} \\ \frac{\partial f}{\partial A_{21}} & \frac{\partial f}{\partial A_{22}} & \cdots & \frac{\partial f}{\partial A_{2m}} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial f}{\partial A_{n1}} & \frac{\partial f}{\partial A_{n2}} & \cdots & \frac{\partial f}{\partial A_{nm}} \end{bmatrix}.$$

Example: The determinant

The *determinant* is a scalar-valued function of a square matrix that we can use to determine whether a matrix is invertible.

The determinant of a $A \in \mathbb{R}^{2\times 2}$ is given by:

$$\det(\mathbf{A}) = |\mathbf{A}| = A_{11}A_{22} - A_{12}A_{21}.$$

For example,

$$\begin{vmatrix} 1 & 2 \\ 3 & 4 \end{vmatrix} = 1 \times 4 - 2 \times 3 = -2$$

Example: The determinant

The determinant of a $A \in \mathbb{R}^{2\times 2}$ is given by:

$$\det(\mathbf{A}) = |\mathbf{A}| = A_{11}A_{22} - A_{12}A_{21}.$$

The gradient of the 2×2 determinent is thus given by:

$$\nabla_{\mathbf{A}} \det(\mathbf{A}) = \begin{bmatrix} \frac{\partial}{\partial A_{11}} \det(\mathbf{A}) & \frac{\partial}{\partial A_{21}} \det(\mathbf{A}) \\ \frac{\partial}{\partial A_{21}} \det(\mathbf{A}) & \frac{\partial}{\partial A_{22}} \det(\mathbf{A}) \end{bmatrix} = \begin{bmatrix} A_{22} & -A_{21} \\ -A_{12} & A_{11} \end{bmatrix}$$

For example,

$$\nabla \begin{vmatrix} 1 & 2 \\ 3 & 4 \end{vmatrix} = \begin{bmatrix} 4 & -3 \\ -2 & 1 \end{bmatrix}$$

The Jacobian matrix

The derivative of a vector-valued function with a vector input, $\mathbf{f} : \mathbb{R}^n \to \mathbb{R}^m$, is called the *Jacobian* matrix:

$$\mathbf{J}_{\mathbf{f}}(\mathbf{x}) = \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \frac{\partial f_1}{\partial x_2} & \cdots & \frac{\partial f_1}{\partial x_n} \\ \frac{\partial f_2}{\partial x_1} & \frac{\partial f_2}{\partial x_2} & \cdots & \frac{\partial f_2}{\partial x_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial f_m}{\partial x_1} & \frac{\partial f_m}{\partial x_2} & \cdots & \frac{\partial f_m}{\partial x_n} \end{bmatrix},$$

where the vector function $\mathbf{f}$ is given by $\begin{bmatrix} f_1 & f_2 & \cdots & f_m \end{bmatrix}$.

You are unlikely to encounter this in CITS5508, however you will see it if you take CITS5017 *Deep Learning* next semester.

Example: polar coordinates

Consider, from geometry, the polar co-ordinate system, where a point in 2D is given by it's length, r and angle from the x-axis, θ.

To convert polar co-ordinates to Cartesian co-ordinates, we use the rules:

$$x = r\cos\theta \quad \text{and} \quad y = r\sin\theta$$

We can write this as a function:

$$f(r,\theta) = \begin{bmatrix} r\cos\theta \\ r\sin\theta \end{bmatrix} = \begin{bmatrix} x \\ y \end{bmatrix}$$

The Jacobian matrix of this function is given by:

$$\mathbf{J_f}(x,y) = \begin{bmatrix} \frac{\partial}{\partial r}(r\cos\theta) & \frac{\partial}{\partial \theta}(r\cos\theta) \\ \frac{\partial}{\partial r}(r\sin\theta) & \frac{\partial}{\partial \theta}(r\sin\theta) \end{bmatrix} = \begin{bmatrix} \cos\theta & -r\sin\theta \\ \sin\theta & r\cos\theta \end{bmatrix}$$

Resources and Further Reading

- Géron's Differential Calculus Tutorial
  - Up to and including "Gradients"
  - Associated with the *Hands-On Machine Learning* textbook
- Boelkins et al.'s *Active Calculus*
  - Chapter 1: Understanding the derivative
  - Chapter 2: Computing Derivatives
  - Sections 1-2 of Chapter 10: Derivatives of Multivariable Functions
- Kolter's Linear Algebra Review course
  - "Course Notes" section 4.1
- Edelman and Johnson's Matrix Calculus for Machine Learning and Beyond course
  - Lecture 1 (Parts 1 and 2), or
  - "Course Notes" chapters 1 and 2
- The Table of Derivatives from Strang and Herman's *Calculus* (OpenStax)
- Wikipedia's Glossary of Mathematical Symbols

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